



MSc Thesis for Health Data Science

A Bayesian Time-Series Framework for Comparing Hospital Pressure Across Irish HSE Health Regions

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Declaration

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Master of Science is entirely our own work and had not been taken from the work of others or Artificial Intelligence, save and to the extent that such work has been cited and acknowledged within the text of our work.

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1 Introduction

2 Data

2.1 Data Collection

3 Methods

3.1 Standardisation

Raw trolley counts vary across regions primarily because of population size rather than differences in hospital pressure. To enable fair comparison, counts were standardised to a rate per 10,000 catchment population.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region catchment population}_i} \times 10,000$$

This transforms the outcome into patients on trolley per 10,000 population per week, removing the confounding effect of region size and allowing direct comparison of hospital pressure across the six HSE regions.

3.2 Model

Let $y_{i,t}$ denote the scaled trolley count for region i at week t , for $i \in \{1, \dots, 6\}$ and $t \in \{1, \dots, T\}$.

Likelihood

$$\begin{aligned} y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1 \end{aligned}$$

Mean function

$$\begin{aligned} \mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + \delta_{\text{pre},i} \cdot \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \cdot \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \cdot \mathbf{1}(t \bmod 52 = 2) \\ &\quad + \sigma_{\text{pre}} \cdot \mathbf{1}(t = 86) \cdot \text{mw}_i + \sigma_{\text{mid}} \cdot \mathbf{1}(t = 87) \cdot \text{mw}_i + \sigma_{\text{post}} \cdot \mathbf{1}(t = 88) \cdot \text{mw}_i \end{aligned}$$

Model V4 mean function

Where $\text{mw}_i = \mathbf{1}(\text{region } i = \text{HSE Mid West})$. The β_i and γ_i coefficients capture the annual seasonal cycle, $\frac{\gamma_i}{\beta_i}$. The $\delta_{\text{pre}/\text{mid}/\text{post},i}$ components are region level new year effects. The σ component captures the Mid West funding reset.

The term $\phi(y_{i,t-1} - \mu_{i,t-1})$ in Equation (3.2) adds an AR(1) component, allowing each weeks observation to depend on the previous week's derivation from the mean.

Priors

The baseline parameters α_i are given a hierarchical prior, treating the six HSE regions as exchangeable within the same national health system:

$$\begin{aligned} \alpha_i &\sim \mathcal{N}(\mu_\alpha, \tau_\alpha^{-1}) \\ \mu_\alpha &\sim \mathcal{N}(0, 0.001) \\ \tau_\alpha &\sim \Gamma(0.001, 0.001) \end{aligned}$$

Remaining priors are independent and uninformative:

$$\begin{aligned} \beta_i, \gamma_i, \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \end{aligned}$$

3.3 Posterior Sampling and Bayesian Ranking

The *Just another Gibbs sampler (rJAGs)*, package was used to draw 20,000 samples from the posterior distribution, A , illustrated below. Health region baselines, α_j^s , were used for ranking. Where j is one of the 6 HSE health regions, $j \in \{1, \dots, 6\}$, and s is the draw index, for $s \in \{1, \dots, n\}$ and total samples is defined as $n = 20,000$.

$$A = \begin{bmatrix} \alpha_1^1 & \cdots & \alpha_j^1 & \cdots & \alpha_6^1 \\ \vdots & & \vdots & & \\ \alpha_1^s & \cdots & \alpha_j^s & \cdots & \alpha_6^s \\ \vdots & & \vdots & & \\ \alpha_1^n & \cdots & \alpha_j^n & \cdots & \alpha_6^n \end{bmatrix}$$

Ranks follow the posterior ranking approach of Laird & Louis (1989), as applied by MacDermott et al. (2025). Where each sample is posterior sample is ranked to form rank distribution, R .

$$R = \begin{bmatrix} r_1^1 & \cdots & r_j^1 & \cdots & r_6^1 \\ \vdots & & \vdots & & \\ r_1^s & \cdots & r_j^s & \cdots & r_6^s \\ \vdots & & \vdots & & \\ r_1^n & \cdots & r_j^n & \cdots & r_6^n \end{bmatrix}$$

Where $r_j^s = \text{rank}(\alpha_j^s) = \sum_{i=1}^6 I(\alpha_j^s > \alpha_i^s)$. Given a tie, ranks are increased by $\frac{1}{2}$ steps for each tied value.

$$r_j^s = \sum_{i=1}^6 [\mathbf{1}(\alpha_j^s > \alpha_i^s) + \frac{1}{2}\mathbf{1}(\alpha_j^s = \alpha_i^s)]$$

Then posterior rank averages, $\hat{r}_i$, are calculated taking the mean across samples.

$$\hat{r}_j = \frac{1}{n} \sum_{s=1}^n r_j^s$$

$$\hat{R} = [\hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6]$$

Then posterior ranked averages are ranked once again to achieve ranked posterior ranked averages are calculated.

$$r_j^* = \text{rank}(\alpha_j^*) = \sum_{i=1}^6 I(\alpha_j^* > \alpha_i^*)$$

$$R^* = [r_1^*, r_2^*, r_3^*, r_4^*, r_5^*, r_6^*]$$

Where $r_j^* = \text{rank}(\hat{r}_j) = \sum_{i=1}^6 I(\hat{r}_j > \hat{r}_i)$.

4 Results

1. DIC comparison
2. Posterior summaries for each model
3. Fitted values plots
4. Regional rankings.

5 Conclusions

In this research...

6 References

- Laird, N. M., & Louis, T. A. (1989). Empirical Bayes ranking methods. *Journal of Educational Statistics*, 14(1), 29–46. <https://doi.org/10.3102/10769986014001029>
- MacDermott, C., Scarrott, C. J., & Ferguson, J. (2025). *Bayes-ically fair: A Bayesian Ranking of the Olympic Medal Table*. <https://doi.org/10.48550/arXiv.2510.14723>

7 Appendix

[Placeholder — full posterior trace plots, Gelman–Rubin statistics, residual diagnostics.]

7.1 Model Inheritance

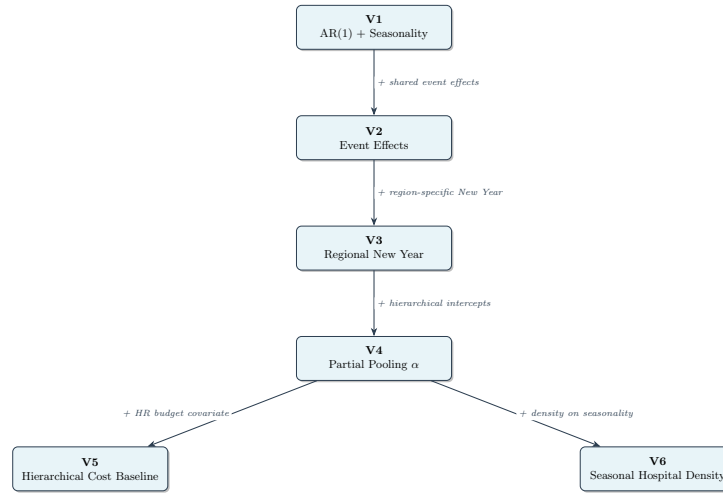


Figure 1: Model inheritance